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Adaptive biases in the wild: Advancing our understanding of the nature of biases

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Abstract

Bias is most often seen as a flaw: people are said to "suffer" from biases and need to be "debiased." Yet a bias, defined simply as a systematic deviation from a norm or standard, can in principle have negative, neutral, or positive meanings and consequences. While the dominant research tradition has emphasized pejorative interpretations by documenting violations of coherence norms such as logic or probability, other perspectives highlight how biases can also be functional and adaptive. This special issue brings together a set of contributions that theorize and provide or discuss empirical evidence on when and why biases may be adaptive. The articles address diverse domains, including framing effects, confirmation bias, reinforcement learning, gambling heuristics, overconfidence, competence attrition in language learning and organizational practices, and the bias blind spot. Together, they show that adaptive biases may not be rare exceptions but a pervasive feature of human cognition and behavior. The articles draw parallels with other pervasive behavioral regularities often conceptualized as biases and lay the foundation for future research on the boundary conditions of the adaptive nature of some biases, the cultural and institutional influences that shape the biases, the interplay between adaptive processes and outcomes, and the emergence and adaptivity of biases at different levels, including individual, team, and organization. By shifting the focus from debiasing to understanding the contextual adaptiveness of biases, this special issue seeks to inspire a more balanced view of bias across psychology and related disciplines.

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1 Introduction

Contemporary academic and lay notions of bias are mostly negative: people "suffer" from biases and need to be "debiased". Indeed, to call someone "biased" is typically meant as a criticism or even accusation. Yet, a bias is, first of all, "just" a systematic deviation from a norm or standard. As such, bias can, in principle, have negative, neutral (i.e., a tendency), and also positive meanings and consequences—depending on the norm and the direction of the deviation from the norm. However, this more nuanced and balanced view of bias has largely been lost in current discourse.

The prevailing negative view of bias stems in large part from lab experimental demonstrations of systematic violations of norms of internal coherence such as logic, probability theory, and expected utility theory. Within this framework, any deviation from the norm is a bias in the pejorative sense because the norm is taken as the gold standard. It is thus by definition impossible in such designs to be biased in a way that has a neutral or positive meaning.

Already decades ago, Herbert Simon urged scholars to expand this perspective on biases. In his reflections on bounded rationality, Simon (2000) emphasized that human cognition is not well described by models of perfect coherence or utility maximization. Instead, he suggested, people rely on simple strategies or shortcuts that reflect both the limits of human cognition and the structure of environments. Thus, what may look like a violation of coherence, may, in fact, be an adaptive solution. As Simon (2000, p. 25) put it: "Rationality is bounded because these abilities are severely limited. Consequently, rational behavior in the real world is as much determined by the 'inner environment' of people's minds … as by the 'outer environment' of the world on which they act." From this perspective, biases need not be dismissed as flaws but understood as regularities that enable decision making under cognitive and informational constraints. In a similar spirit, Viale (2000) distinguishes between biases as errors and as functional enablers. For instance, he describes the "Cartesian bias" of equating cognition with deliberate reasoning while neglecting unconscious mechanisms, and notes that individuals are often aware of the outcomes of their reasoning but not of how they reason. What appears as bias may thus be less about error than about the hidden architecture of cognition.

The idea that biases can be adaptive is more common in, but not limited to, an evolutionary perspective: numerous biases—in the sense of tendencies to attend, perceive, infer, learn, feel, act—have evolved as efficient and sometimes life-saving adaptations to the environment. For example, humans have a strong bias to feel disgusted by food that they got sick from, much more than they have a tendency to love food that was good for them. A bias for sure, but a very useful one. Biases can also be adaptive to our human-made environment. For example, in most countries, the legal system is, for good reasons, quite heavily biased to assume innocence until proven guilty.

We edited this special issue to provide space for the exploration of adaptive biases. Given the journal pages devoted to negative depictions of biases, this may appear like a drop of water in the ocean of the bias literature. However, we hope that like a drop dilutes and spreads through the ocean, the papers in this special issue make the kind of interesting, important, thought-provoking, and perhaps even provocative contribu-

tions that will impact how biases are seen not only in psychology but in disciplines such as management, leadership, organizational studies, political science, and others.

We challenged scholars to submit papers to this special issue that address important research questions in the domain of adaptive biases. We particularly sought manuscripts that shed light on important ongoing debates, examine adaptive biases in real-world contexts such as management, organizations, leadership, law, or society in general, or open up promising directions for future research. We welcomed papers taking different approaches to adaptiveness, featuring quantitative and qualitative studies, inductive and deductive approaches, and empirical as well as conceptual contributions. Overall, we believe that, as a set, the papers comprising this special issue meet the challenge we posed.

2 Overview of contributions

In the invited opening article, Gigerenzer (2025) provides a historical perspective on the notion of bias. Specifically, he distinguishes between two historical views on bias in psychology. First, bias has been viewed as functional and indeed necessary for cognition. Second, bias has been viewed as an error—interfering with cognition—that should be avoided. Further, he points out that in the functional perspective, the notion of bias is value-neutral, neither good nor bad. Rather, the question is under what conditions and for what purposes bias is more or less functional. A key conclusion is that certain biases are not only functional but necessary for cognition to function. From the error perspective, the notion of bias is value-laden: bias is bad. Further, bias is bad because it violates certain norms, such as a true state, expected utility theory, or Bayes' theorem. In contrast, from a functional perspective, the issue is not truth, but getting things done, and as long as biases can "get things done", they are desirable.

In their article on when and why framing effects should not be regarded as errors or mistakes, McKenzie et al. (2025) argue that framing effects are not necessarily irrational and in fact might be adaptive both in laboratory settings and natural environments. Framing effects refer to violations of the normative coherence principle of description invariance and play a central role in the debate regarding human rationality. The authors suggest that for framing effects to be considered *errors* in laboratory settings, the implicit assumption behind description invariance that rational preferences are complete should hold. However, people do not necessarily have a resolved, complete ordering of all possible options and acts. Moreover, frames in typical attribute framing tasks are often not information-equivalent, and because a speaker's choice of frames "leaks" choice-relevant information, *failing to exhibit* a framing effect may be irrational. For the same reasons of preference incompleteness and the informational value of frames, framing effects may not be *mistakes* in natural environments either. Readers are also invited to consider how similar lessons may extend to other coherence norms, such as procedure invariance and independence of irrelevant alternatives.

Hahn's (2025) contribution focuses on one of the most famous biases of all, a bias that has not only been the topic of numerous research studies but also made it into

popular discourse: the confirmation bias. In her paper on confirmation bias, Hahn discusses different notions of confirmation bias, distinguishing broadly between confirmation bias in evidence search, evidence selection, evidence evaluation, and evidence communication. Beyond the categorization of confirmation biases, Hahn proposes a rational basis for the apparent bias in evidence search that stems from Bayesian expectations about the real-world distribution of evidence supporting or contradicting a claim. In essence, Hahn takes the structure of the environment and people's beliefs as central in evidence search, reflecting Simon's perspective on bounded rationality.

Palminteri (2025) examines the relationship between human reinforcement learning processes (i.e., learning from rewards and punishments) and biases, with a focus on the computational characterization of biases and possible applications to behavioral public policy. The author proposes a taxonomy for evaluating biases within the reinforcement learning framework and distinguishes between epistemic biases (errors in internal representations of option values) and praxic biases (errors in action selection). He specifically highlights two pervasive epistemic biases, namely context-dependent outcome encoding (e.g., range normalization) and positivity bias in option value updating. Drawing on computational models, behavioral evidence, and recent work with large language models (LLMs), he argues that these biases, while sometimes detrimental in laboratory settings with respect to objective accuracy (i.e., unbiased knowledge), may provide advantages in a wide array of naturalistic environments by enhancing instrumental accuracy (i.e., better outcomes). Readers are invited to consider how relative value encoding in the form of range normalization and a moderate insensitivity to negative outcomes can sometimes be advantageous. Yet, they are also cautioned against concluding that the biases are "inherently beneficial" and, in a truly Brunswikian tradition (Brunswik 1952, 1955), reminded of the importance of the match between a given bias and the specific decision environment.

Bennis (2025) takes us on a fascinating journey to the casino, exploring the use of heuristics among casino blackjack players. Relying on extensive field observations and quantitative data, he documents how blackjack players adopt a simple heuristic to decide when to take additional cards, the most common blackjack decision. The heuristic requires them to make a clearly wrong assumption (i.e., that all upcoming cards will be 10-value cards, which is true less than one-third of the time). Yet, while leading to systematic error, the heuristic is adaptive in the sense that it both provides an easier alternative to the optimal strategy and results in better expected returns at all levels of playing experience. Beyond the specific results on what appears to be an adaptive bias, the author engages the reader in a broader discussion of whether gambling is a good "domain-general metaphor for decision making under risk or uncertainty" and what models and rules may most accurately represent people's decisions in domains where outcome likelihoods and monetary outcomes are quantifiable and unambiguous.

Page (2025) focuses on another famous bias: overconfidence. His paper explores the strategic benefits of overconfidence. He begins by pointing out that people's self-serving tendencies in their beliefs—such as believing that they are better than they actually are—has often been interpreted as evidence that they are bad at forming judgments about themselves and the world. In contrast, Page argues that these biased

beliefs are due to evolution selecting for beliefs that are useful (rather than true). Applied to the context of his paper, "overconfidence may actually be an adaptive feature of our cognition, providing advantages in various situations." He further argues that in addition to the individual benefits, overconfidence can generate collective costs and, when becoming widespread, it may "lead to inefficient social equilibria—such as bargaining impasses, costly conflicts, or misguided leadership."

Guercini and Lechner (2025) examine borrowing in competence attrition as an adaptive bias. To make their case, they draw on research on language attrition in linguistics. Language attrition is loss of proficiency in one's native language due to interaction with and learning of a new language. Extending this idea to organizational contexts, they argue that borrowing, as a specific form of competence attrition, is a "heuristic solutions to a problem of partial loss of the pre-existing organizational competences." With the help of a case study of an Italian company in the sneakers industry, they show how "resorting to borrowing represents a bias in standard skills that can be viewed positively by the organization as an adaptive solution, although not always successful." They discuss the potential of the notion of competence attrition to further our understanding of how adaptive biases can support organizational learning and performance.

Finally, Cruz and Mata (2025) examine how people perceive their own biases, distinguishing between those seen as negative and those seen as positive. They provide evidence for a "motivated bias blind spot": people are more likely to acknowledge a bias in themselves when it is described in terms of its positive outcomes but are less likely to do so when the bias is associated to negative consequences. This suggests that certain biases may be regarding as socially desirable, and thus adaptive, even if they can be seen as logical errors.

3 Future work and conclusion

Overall, the contributions in the special issue show not only that adaptive biases are widespread but also examine what makes these biases adaptive and when they are adaptive. Yet, many questions remain for future research. For example, what are the boundary conditions that determine whether a bias proves adaptive or maladaptive? What other biases can be adaptive at the individual level but harmful at the collective or societal level, and do people recognize potentially contradictory consequences of some biases? Beyond evolutionary explanations, what roles do culture and human-made environments, such as legal, political, social, or economic institutions, play in shaping biases and their potential adaptiveness? And how should we think about the relationship between heuristics and adaptive biases: Are biases necessarily products of evolved heuristics, or can they also arise from deliberate analysis?

Future work is also needed to clarify the distinction between biased processed and biased outcomes. Can an adaptively biased process lead to maladaptive outcomes? And can an adaptive outcome result from a maladaptive process? Moreover, most existing research has tended to examine adaptive biases at the level of individuals or agent. What about adaptive biases that only emerge at the level of teams or organizations? Here, particularly interesting would be examinations of biases that emerge

only at the team or organizational level, driven by individual, team, or organizational processes. Finally, further advances in methodology, conceptualization, and practical applications are needed to refine how we define and measure adaptive biases, to understand their antecedents and consequences, and explore how adaptive biases may be encouraged, taught, or implemented.

We hope that the articles in this special issue will inspire renewed interest in the nature of biases in human cognition, and in their potential adaptive role across diverse contexts. We look forward to future research that continues to explore when, why, and how biases may—or may not—be adaptive and functional.

References

Bennis WM (2025) A heuristic based on transparently false likelihoods improves gamblers' expected value in the wild. Mind Soc. https://doi.org/10.1007/s11299-025-00346-9

Brunswik E (1952) The conceptual framework of psychology. University of Chicago Press

Brunswik E (1955) Representative design and probabilistic theory in a functional psychology. Psychol Rev 62(3):193–217

Cruz F, Mata A (2025) Motivated bias blind spot: people confess to more or less bias depending on its desirability. Mind Soc. https://doi.org/10.1007/s11299-025-00325-0

Gigerenzer G (2025) Two kinds of bias. Mind Soc. https://doi.org/10.1007/s11299-025-00362-9

Guercini S, Lechner C (2025) Borrowing in competence attrition as adaptive bias. Mind Soc. https://doi.org/10.1007/s11299-025-00341-0

Hahn U (2025) Confirmation bias and the plausible distribution of evidence. Mind Soc. https://doi.org/10.1007/s11299-025-00360-x

McKenzie CRM, Sher S, Liu XS et al (2025) When and why framing effects are neither errors nor mistakes. Mind Soc. https://doi.org/10.1007/s11299-025-00338-9

Page L (2025) The strategic benefits of overconfidence. Mind Soc. https://doi.org/10.1007/s11299-025-00355-8

Palminteri S (2025) Human reinforcement learning processes and biases: computational characterization and possible applications to behavioral public policy. Mind Soc. https://doi.org/10.1007/s11299-025-00329-w

Simon HA (2000) Bounded rationality in social science: today and tomorrow. Mind Soc 1(1):25–39

Viale R (2000) Biases and fallacies in reasoning: the mind–society problem. Mind Soc 1(1):5–24

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